
Chapter 1: A Square and a Cube

Worksheet 1: Squares, Cubes, Square Roots, and Cube Roots

Learning Objective: Students will find squares and cubes of numbers; identify perfect squares and perfect cubes; compute square roots and cube roots using prime factorisation and estimation; apply these concepts to real-life problems.

Worked Example

Find the square root of 1764 by prime factorisation.

Solution: $1764 = 2^2 \times 3^2 \times 7^2$

$$\sqrt{1764} = 2 \times 3 \times 7 = 42$$

Find the cube root of 13824.

$13824 = 2^9 \times 3^3 \times \dots$ Actually $13824 = 24^3$ since $24^3 = 13824$.

$$\therefore \sqrt[3]{13824} = 24$$

Questions 1–5: Easy / Recall

Q1. Find: 15^2 , 23^2 , $(-7)^2$.

Q2. Find: 4^3 , 10^3 , $(-3)^3$.

Q3. Is 225 a perfect square? If yes, find its square root.

Q4. Is 512 a perfect cube? If yes, find its cube root.

Q5. Write all perfect squares between 100 and 200.

Questions 6–10: Basic Application

Q6. Find $\sqrt{2916}$ using prime factorisation.

Q7. Find $\sqrt[3]{9261}$ using prime factorisation.

Q8. A square floor has area 1444 sq. m. Find the side length.

Q9. How many rows and columns can 729 students stand in for a square formation?

Q10. Without calculating, determine whether $\sqrt{7921}$ is odd or even. (Hint: look at the last digit.)

Questions 11–15: Practice and Skill Building

Q11. Find the smallest number by which 720 must be multiplied to make it a perfect square.

Q12. Find the smallest number by which 1944 must be divided to make it a perfect cube.

Q13. Evaluate: $\sqrt{6.25}$ and $\sqrt{0.0081}$.

Q14. Find $\sqrt[3]{0.001331}$.

Q15. If $x^2 = 12.96$, find x .

Questions 16–18: Higher Order Thinking

Q16. The sum of first n odd natural numbers equals n^2 . Use this to find: (a) $1 + 3 + 5 + \dots + 19$ and (b) how many odd numbers must be added to get a sum of 169.

Q17. A cubical water tank holds 15,625 litres. Each litre = 1000 cubic cm. Find the side of the cube in cm.

Q18. The difference of squares of two consecutive even numbers is 100. Find the numbers. (Hint: $(n + 2)^2 - n^2 = 100$.)

Questions 19–20: Real-Life Applications

Q19. A school wants to plant trees in a square grid. They have 1000 saplings. What is the largest perfect-square number of saplings they can plant? How many are left over?

- Q20.** A cuboid-shaped box has volume 2744 cubic cm and is a perfect cube shape. Find its edge length. If you stack 8 such boxes into a larger cube, what is the side of the larger cube?

Reflection Question: Is the square of every odd number also odd? Is the cube of every even number also even? Justify with examples.